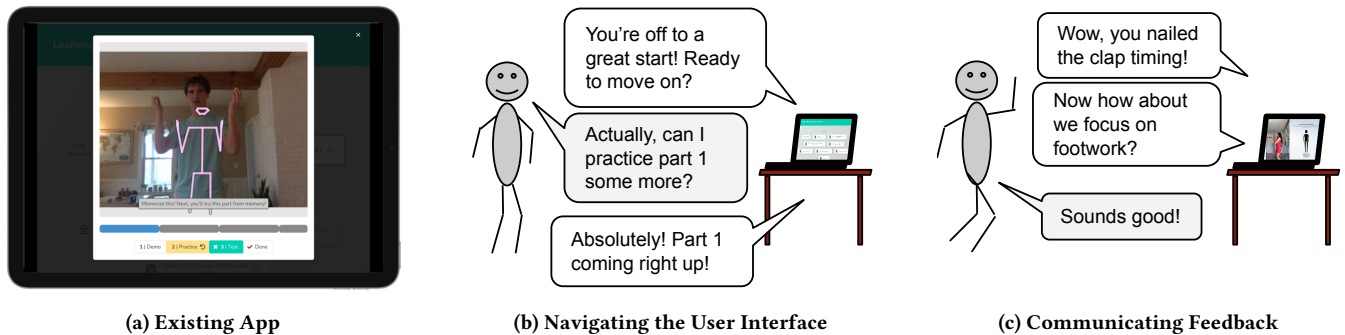


# Integrating a LLM into an Automatic Dance Practice Support System: Breathing Life Into the Virtual Coach

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**Figure 1: Proposed System.** Integrates a LLM into our existing dance learning app to create a handsfree “Virtual Dance Coach” experience.

## ABSTRACT

We propose using a LLM to breathe life into our existing dance learning app in the form of an AI dance coach persona. The current app guides the user through dance practice plans, using the webcam and pose estimation to give feedback. Using the LLM, voice recognition, speech synthesis, and affect recognition, we plan to transform the interface from a mechanical click-on-screen experience to a hands-free speak-with-the-coach interaction. In particular, we’ll use the LLM to announce coaching guidance, provide encouragement, communicate feedback, and interpret the user’s spoken intent.

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## 1 INTRODUCTION

Don’t you wish you could dance? Dance choreographies are challenging to learn, yet fun and freeing to perform. We propose developing an AI dance coach to help you express your inner rhythm. Our existing dance teaching app [2] supports dance practice by breaking down a dance into smaller parts, creating a lesson plan,

and (new for UIST this year [3]) using pose-estimation and your device’s webcam to compute a score and offer feedback as you go.

Using LLMs, we propose to breathe life into the dance learning experience, transforming it from a web tool into a dance coach. Given motion descriptors and computed comparison metrics, we will use the language model to communicate feedback in an intuitive and human-like way. The language model can also be used to process user input—for example, the software may ask the users “what do you need more help with?”, and the LLM can process the answer, along with other context such as dance structure and practice history, to generate a customized practice activity.

Dancing is an embodied experience; a process that develops the connection between our bodies and minds; an activity that grounds us in the physical space that we live in. Using a LLM in combination with microphone input, voice recognition, speech synthesis, speaker output, and augmented with context from affect recognition and pose estimation, we want to “de-screen” the interface—moving the focus of attention away from the computer and towards the dancer’s own body and physical environment.

Dance is also a social activity. Energy is exchanged between humans when they move together; a physical-spiritual-emotional connection is made. This energy exchange is lacking when practicing in front of a YouTube video, or with the support of a computer. By creating a persona for the computer, we hope to inject energy into the practice process, elevating from a drill into a pleasurable, meaningful activity.

## 2 LLM TASKS & INTEGRATION PLAN

We plan to integrate the LLM into our app in four places, as we describe below. We will leverage the long context support of the Claude API by providing extensive contextual information about the user’s performance metrics, estimated emotional state, activity

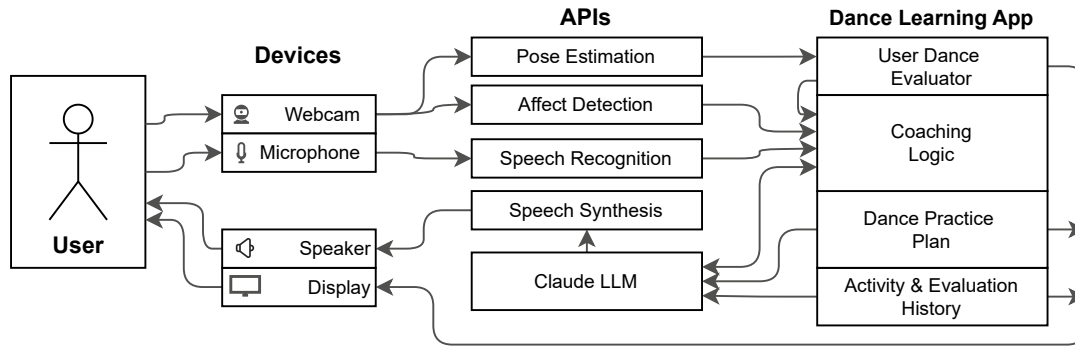
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**Figure 2: Schematic of LLM Integration.** For each of the tasks, the LLM will be supplied with relevant context from the app and asked to generate empathetic, adaptive output.

history, the nature of the dance, the organization of the practice plan, and the current practice activity, as documented in Fig. 2.

**Task 1** (Drive a Spoken User Interface). Learning a dance choreography requires practicing the movements in open space, so it is disruptive to the experience for the user to move to a device in order to click a mouse, touch a screen, or read detailed text. We plan to use the Claude API to power a spoken dance-coaching interface, providing context and giving prompt direction to provide varied, individualized responses.

**Subtask 1.1** (Announce Navigation Options). Generally, we expect users to follow a similar overall learning pattern: practicing one section at a time and incrementally integrating the new section with previous ones. However, there is considerable variation in the amount of practice that individuals will need on each section, as well as personal preferences in the learning order. To accommodate this, we will create a set of options (such continue, repeat, repeat slower, skip, etc) for the user to choose from and use the Claude API announce these options. Additionally, we will experiment with using Claude API to rank these options and provide an intelligent suggestion based on the current context, and based on the user’s emotional needs (as described in the feedback section). We will then use the Web Speech API to synthesize and play the announcement.

**Subtask 1.2** (Interpret User’s Spoken Intent). The other side of a spoken interface is to interpret the user’s speech. Using the Web Speech API, we’ll convert the user’s speech into text, then feed it into the Claude API as part of a templated prompt, asking Claude to identify which of the navigation options the user desires (with a fallback if there isn’t a clear intention).

**Task 2** (Communicating Feedback). Perhaps the most interesting of the tasks we set out for the LLM is that of communicating feedback to the user. The current app performs pose-estimation on the user’s webcam feed using the Mediapipe [5] framework, which is then compared to the poses of the dancer in the source video. We can compute a variety of motion descriptors [4] based on the kinematic information of the user and reference dances. Through this, we can derive a detailed picture of the user’s current and historical performance; however it can be difficult to distill a package of technical metrics into concise, useful feedback for the user. We

will address this feedback challenge by providing the Claude LLM with this computed motion context and guiding it to be deliberative in the feedback it provides through the use of a problem-solving template and the tree-of-thought technique [7].

**Subtask 2.1** (Practice Recommendations). As part of feedback communication, we'll use the LLM to generate practice activities that target periods of a dance where the user struggles.

**Subtask 2.2** (Empathetic Responsiveness). As an additional twist on the feedback generation, we will prompt Claude to incorporate the user’s emotional needs into its decision making. As put forth in Wulf and Lewthwaite [6], motivation is key to effective motor learning. By providing the Claude LLM with performance history and facial affect context (using [1]), we’ll enable our AI Dance Coach to responsively adapt its feedback – softening or withholding critical feedback when advantageous, or becoming more frank and demanding when the user has the capacity.

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